

Muhammad Roshaan

Computer Science Researcher & Engineer

Muhammadroshaan1127@gmail.com | +92 315 824 6881 | linkedin.com/in/muhammad-roshaan-024ab2288
github.com/MRoshaan | m-roshaan.me | Karachi, Pakistan

Summary

Final-year BSCS student (CGPA 3.80/4.00, SSUET) specializing in scientific LLM evaluation, AI safety, and Python/Go engineering. First author of an accepted SLM-Agents paper and co-author of an accepted AI-WiLD paper (NeurIPS 2026).

Research & Publications

Agent Watchtower – Order-Invariant Causal Trace Verification Aug 2026

Primary author. Verify-Agents and AI-WiLD Workshops @ NeurIPS 2026 (under review).

- Built a Go causal-DAG renderer and 160-cell serialization sweep: deterministic verdicts had a 0.000 flip rate, while live LLM judges flipped on 55–75% of identical traces (kappa 0.08–0.29; Gemini, DeepSeek, Kimi).

BitFracture – Quantization Error Taxonomy for Small Tool-Calling Models 2026

Primary author. Accepted (poster), SLM-Agents Workshop @ NeurIPS 2026.

- Ran a pre-registered FP16-vs-NF4 study with a 7-class error taxonomy and bootstrap 95% CIs: 4-bit NF4 nearly tripled missed required tool calls for a 1.7B model (6.0% → 17.3%) while a 4B model showed a powered null.

SafetyShield – Bayesian-Efficient AI-Safety Evaluation and Guardrail (Final Year Project) 2026–2027

Team of 4, SSUET. Supervisor: Dr. Nadeem.

- Designed a Bayesian early-stopping evaluation engine that detects risky model behavior with far fewer prompt evaluations, plus a probe dataset, guardrail kit, and examiner dashboard; explained the architecture to the team.

CanaryGame – Shared-Memory Honeytokens for Multi-Agent LLM Systems 2026

Contributing author. Accepted (poster), AI-WiLD Workshop @ NeurIPS 2026.

- Measured zero post-trigger harm (95% CI [0, 0]) and 0.80 trap activation with brokered containment in a local multi-agent simulation; shared-memory coalitions pooled decoy fingerprints and increased harm over baseline.

Reviewer, Verify-Agents Workshop @ NeurIPS 2026

Accepted Aug 2026

Experience

AI/LLM Evaluator Sep 2026 – Present

Scientific AI Evaluation – Physics & Materials Science.

- Analyze Physics and Materials Science papers to design LLM benchmarks requiring scientific reasoning and computation.
- Develop Python/Jupyter Notebook prompts, reference solutions, and executable tests for numerical accuracy and reproducibility.
- Benchmark Google Gemini, OpenAI GPT, and Anthropic Claude; analyze repeated-run failures and refine tasks and tests.

Data Science Intern – 10Pearls Pakistan Apr 2026 – Jun 2026

10Pearls Shine Internship Program. Remote.

- Built and evaluated machine learning workflows covering data preparation, feature engineering, model evaluation, and experiment tracking (Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Hopsworks).

Selected Engineering Projects

SeatVault – Transaction-Safe Concurrency Engine 2026

- Combined transaction boundaries, PostgreSQL row locks, and Redis distributed locking to prevent overselling under concurrent reservations, with idempotent background workflows (Python, FastAPI, Celery, Docker).

Sentinel – Real-Time Fraud Detection with Explainable AI 2026

- Built a real-time fraud-detection service covering feature preparation, model serving, explainability, and operational monitoring (Python, FastAPI, scikit-learn, XGBoost, LightGBM, Redis, PostgreSQL).

BooknScore – Offline-First Scoring and AI Service 2025–2026

- Designed zero-data-loss offline sync between SQLite and Supabase/PostgreSQL, plus a Python/Gemini multimodal service converting match data into commentary and production scripts (Flutter, LLM agents).

Technical Skills

Languages & Systems: Python, Go, Java, FastAPI, Spring Boot, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, Celery, Docker

ML & LLM Evaluation: scikit-learn, XGBoost, LightGBM, Pandas, NumPy, LLM evaluation, tool calling, fault injection, guardrails, Bayesian statistics, Gemini, vLLM

Engineering Tooling: pytest, GitHub Actions, OpenTelemetry, Jupyter, Git, Power BI, DAX

Education

Sir Syed University of Engineering and Technology Sep 2023 – Jul 2027

BS in Computer Science | Final Year, 7th Semester | CGPA: 3.80/4.00

Relevant coursework: AI, Machine Learning, Data Structures & Algorithms, Databases, Operating Systems, Software Engineering, Networks